

PRO

IOMINDS.AI · DETAILED CURRICULUM

Pro — Innovators

Advanced level · Ages 13–15 · 48 sessions over 6 months · full-stack IoT, edge AI and competition-ready capstones

Level

Advanced

Ages

13–15

Sessions

48

Capstone

Deployable



IOMinds.ai

<internet of minds>

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Course overview

Pro is the **Innovators** track — full-stack IoT and edge AI for teenagers ready to take an original idea from breadboard to competition stage. Students work with the same tools professional engineers use, document their work like engineers, and graduate with a deployable capstone innovation.

Level	Pro — Advanced
Ages	13–15 years
Duration	48 sessions over 6 months
Focus	Full-stack IoT · edge AI · PCB design basics · data pipelines · project documentation
Tools	Python, Raspberry Pi, ESP32, TensorFlow Lite, MQTT, Node-RED, Fritzing, GitHub
Capstone	A deployable innovation ready for ATL Marathon, WRO, or Codeavours

Module 1 — Engineering-grade Python (Sessions 1–8)

S1–S3	Python, properly	Functions, modules, files and error handling; reading documentation; version control with GitHub from day one.
S4–S6	Data pipelines	Collecting, cleaning and charting sensor data; CSVs, JSON and simple databases.
S7–S8	Mini-build	AI weather predictor: a month of local data trains a model that predicts tomorrow's temperature.

Module 2 — Full-stack IoT (Sessions 9–20)

S9–S11	Raspberry Pi	Linux basics, GPIO, and the Pi camera; when to choose a Pi vs. a microcontroller.
S12–S14	MQTT & Node-RED	Publish/subscribe messaging between devices; visual flows wiring sensors to dashboards to actions.
S15–S17	Smart home stack	ESP32 controls LEDs and fans via smartphone and voice — a complete home-automation slice.
S18–S20	Air quality system	MQ-series sensors monitor indoor air and trigger multi-channel alerts when pollution rises.

Module 3 — Edge AI & computer vision (Sessions 21–32)

S21–S23	Neural nets, gently	Handwritten digit recognition (0–9) — layers, weights and training loops made visible.
S24–S26	TensorFlow Lite	Running trained models on-device: why the "edge" matters for speed and privacy.
S27–S29	Vision projects	Pi-camera surveillance system with detection; emotion detector reading happy/sad/angry expressions.
S30–S32	AI + IoT fusion	A model's decision drives real hardware end-to-end; latency, reliability and failure handling.

Module 4 — From prototype to product (Sessions 33–40)

S33–S35	PCB design basics	From breadboard to Fritzing schematic to a manufacturable board layout.
S36–S37	Enclosures & power	Batteries, chargers and cases; making a build survive outside the lab.
S38–S40	Documentation	READMEs, wiring diagrams, demo videos and pitch decks — communicating like an engineer.

Module 5 — Capstone & competition (Sessions 41–48)

S41–S44	Capstone build	A unique IoT + AI innovation designed and coded entirely by the student, with mentor reviews each week.
S45–S46	Competition prep	Rules, judging criteria and rehearsals for ATL Marathon, WRO and Codeavour.
S47–S48	Grand showcase	Public demo day; portfolio finalised on GitHub; graduation with the capstone project.

Every enrolment includes

Hardware kit included with the course · lifetime access to study material · lifetime support for queries · lifetime updates on national and international competitions · course completion certificate.

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